29 December 2025

Drop Noise For Cleaning LLMs Data

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Abstract

In industry NLP application, our dataset by prompting large language models (LLMs) has a certain number of noise data. We present a simple method to find the noise data and remove them. We retain the data that contains certain common tokens between the LLMs data and the prediction results of a generative model trained on the LLMs data. We remove the data that does not contain certain common tokens between the LLMs data and the prediction results of a generative model trained on the LLMs data. We adopt T5-Base as our generative model. The experiment result shows our method is highly effective and does not require any manual annotation for text generation task. For industry deep learning application, our method improves the text generation task's accuracy from 88% to 98% under human evaluation, meanwhile the LLMs data source is sufficiently abundant. However, for classification tasks, simply dropping noisy data does not lead to any improvement.

Keywords

computing and processing, data cleaning, deep learning, large language models

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Drop Noise For Cleaning LLMs Data

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Abstract. In industry NLP application, our dataset by prompting large language models (LLMs) has a certain number of noise data. We present a simple method to find the noise data and remove them. We retain the data that contains certain common tokens between the LLMs data and the prediction results of a generative model trained on the LLMs data. We remove the data that does not contain certain common tokens between the LLMs data and the prediction results of a generative model trained on the LLMs data. We adopt T5-Base as our generative model. The experiment result shows our method is highly effective and does not require any manual annotation for text generation task. For industry deep learning application, our method improves the text generation task's accuracy from 88% to 98% under human evaluation, meanwhile the LLMs data source is sufficiently abundant. However, for classification tasks, simply dropping noisy data does not lead to any improvement.

Keywords: Deep Learning · Data Cleaning · LLMs

1 Introduction

With the development of LLMs [1–3], prompting LLMs to obtain a domain-specific training dataset and then training a smaller model to achieve sufficiently fast model inference has become a very useful approach for performing NLP tasks. However, the accuracy of LLMs' original datasets within a specific domain generally only reaches 88%-90%. Additionally, using manual annotation methods to correct or clean the datasets requires a substantial amount of human resources and time. Therefore, this paper proposes a method for automatic dataset cleaning. Experiments show that it is highly effective, requires no manual annotation, and can be extended to more AI tasks based on LLMs datasets, such as computer vision.

2 Method

The whole pipeline and algorithm pseudocode of our method are shown in Figure 1 and Algorithm 1. We conducted experiments on a NLP task that inputting item names to generate item tags. We prompted the LLMs to write synonyms for the item name that can be found through search.

In our task, the T5-Base model we trained overfitted on a dataset with a size of 50,000. We trained for 30-50 epochs. Our threshold condition is that the number of common tokens is greater than 20% of the total length of the two text data.

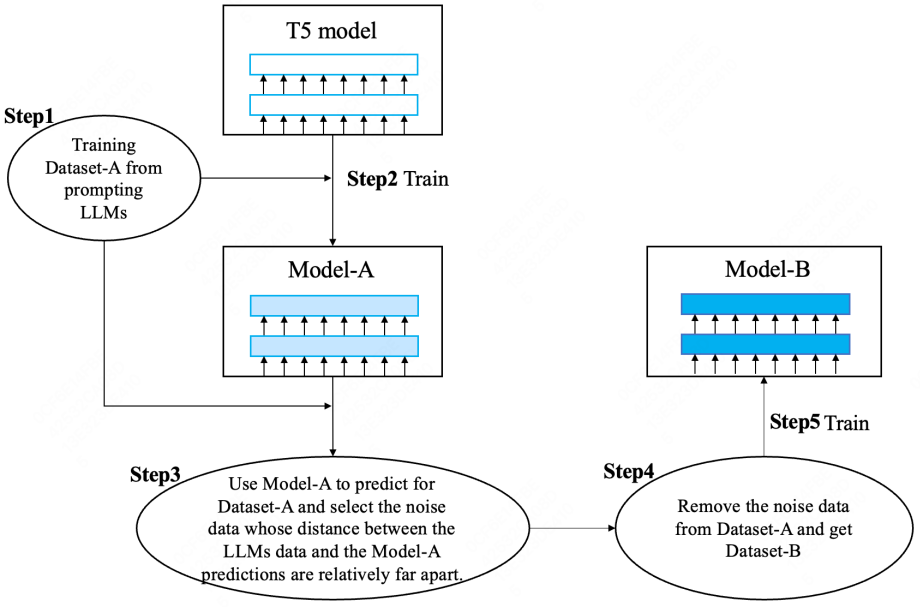


Fig. 1. Remove the noise data

Algorithm 1 Self-Drop To Clean LLMs Dataset

Input: Dataset datasetV1 by prompting LLMs.
Output: Dataset datasetV2.

```
datasetV2 = [ ]
modelV1 = train_on(datasetV1)
for oneLlmData <- datasetV1 do
  output = modelV1.predict(oneLlmData.input)
  if common_token_score(output, oneLlmData.output) > THRESHOLD then
    datasetV2.append(oneLlmData)
  end if
end for
return datasetV2
```

3 Experiments

We conducted experiments on a NLP task that inputting item names to generate item tags. We prompted the LLMs to write synonyms for the item name that can be found through search. We constructed an initial dataset of 50,000, which was reduced to 35,000 after being cleaned using our method. The results of the experiment are shown in Table 2.

Table 1. The experiment results of text generation by T5-Base. Model-A, Model-B are corresponding to Fig 1.

Model In Fig 1	Human Evaluation Accuracy
Model-A	88.0%
Model-B	98.0%

Table 2. The experiment results of text classification by pretrained BERT. Model-A, Model-B are corresponding to Fig 1.

Model In Fig 1	Dev Dataset Accuracy	Train/Dev Dataset Size
Model-A	77.6%	20928/1152
Model-B	96.4%	20074/894 (train/dev-part dropped)
Model-B	77.6%	20074/1152 (train-part dropped)

4 Discussion

In actual observations for Table 1 (text generation task), we found that almost all noisy data with significant issues were filtered out. The significant issues here refer to the presence of unexpected special tokens in the data. This automated cleaning method is a highly compatible combination with the method of obtaining data through prompting LLMs. The amount of training data obtained via prompting LLMs is sufficient, and we can set the necessary control thresholds to filter the data.

But at Table 2 (text classification task), it demonstrates that while LLM-annotated data may contain some noise, the method proposed in this paper can effectively remove such noise. However, this process does not alter the fundamental "LLM annotation rules", "LLM knowledge sources for annotation", or "LLM annotation biases" inherent in the LLM-annotated data.

4.1 Why drop-noise method work?

Why drop-noise method work? Because deep learning is statistic-based. Take classification as example. (In a broad sense, all the machine learning tasks can be viewed as classification.)

If there are three very similar data (data-1/data-2/data-3) in total, which labels are class-A/class-A/class-B, Then the trained model will probably predict class-A for data-3.

We suppose that data-3 is wrong-labeled by human, because more people labeled these very similar data-1/data-2 to class-A.

And the trained model predict class-A for data-3. So the noise data here is data-3 by our method.

If we do not drop data-3, the model prediction for new data that is the most similar to data-3 will be class-B, which is wrong. The new data is more similar to data-3 than data-1/data-2.

If we drop data-3, the model prediction for new data that is the most similar to data-3 will be class-A, which is right.

4.2 Comparing to RAG methods

Our approach is to obtain a dataset from LLMs to train our own model, in contrast to the RAG method [6], which involves working on the prompts given to LLMs.

For high-precision deep learning tasks, another approach is the RAG (Retrieval Augmented Generation) method, which involves providing more information to the prompts for LLMs. Upon observation, the issue with the RAG-LLM method is that some results returned by the LLMs are still uncontrollable and can be considered as noise for specific tasks.

The key distinction between the RAG approach and data cleaning methods lies in the fact that the knowledge sources or additional information provided to LLMs in RAG inherently contain noisy data, which limits the upper bound of the final LLM outcomes.

5 Relate Work

T5 [4] is a model based on the transformer encoder-decoder architecture [5]. In our experiments, we found that it performs better than the decoder-only GPT in single-turn question-answer style NLP (natural language processing) tasks.

6 Conclusion and Future Work

This paper proposes an automated, annotation-free method for cleaning dataset obtained from LLMs through prompting. In human evaluations, this method can increase accuracy from 88% to 98% in our NLP tasks. Additionally, this method can also be extended to other LLMs data-driven tasks, such as computer vision tasks.

Since we verify that using the methodology in this paper to clean LLM data enables effective training of a smaller model, applying this same approach to clean the complete post-training dataset originally used for training the LLM will become the direction we are to explore.

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